

Deep Generative Models

Lecture 8

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Recap of Previous Lecture

Forward Gaussian Diffusion Process

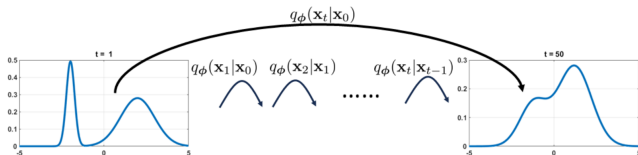
Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$, $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, \mathbf{I});$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

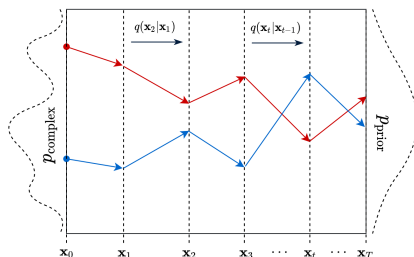
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}).$$



Recap of Previous Lecture

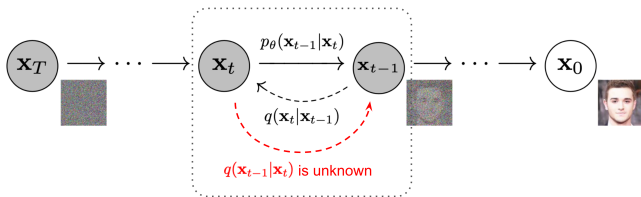
Diffusion describes the process where particles migrate from regions of high density to regions of low density.



1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$;
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$, with $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, $t \geq 1$;
3. $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$, for $T \gg 1$.

If we can invert this process, we would have a way to sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ using noise samples, i.e. $p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$. Hence, our objective becomes to reverse this process.

Recap of Previous Lecture



Reverse Process (Ancestral Sampling)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

The Feller theorem guarantees this approximation is valid.

Forward Process

1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$;
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$;
3. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$.

Reverse Process

1. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$;
2. $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$;
3. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.

Recap of Previous Lecture

Forward Process: Converts an arbitrary distribution $p_{\text{data}}(\mathbf{x})$ into the standard normal $\mathcal{N}(0, \mathbf{I})$ by incrementally adding noise.

$$\begin{aligned}q(\mathbf{x}_t | \mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}); \\q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}).\end{aligned}$$

Reverse Process: This intractable distribution can be well-approximated by a Gaussian (with unknown parameters) when β_t is sufficiently small.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

Conditioned Reverse Process: This Gaussian, with known parameters, describes how to denoise a noisy image $\mathbf{x}_t$ when the final clean image $\mathbf{x}_0$ is known.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

Recap of Previous Lecture

- ▶ $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$ represents the latent variables.
- ▶ Variational posterior distribution:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

- ▶ Generative model and prior:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = p_{\theta}(\mathbf{x}_0|\mathbf{x}_1); \quad p_{\theta}(\mathbf{z}) = \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T)$$

ELBO

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\phi, \theta}(\mathbf{x}) \rightarrow \max_{q, \theta}$$

$$\begin{aligned} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t} \end{aligned}$$

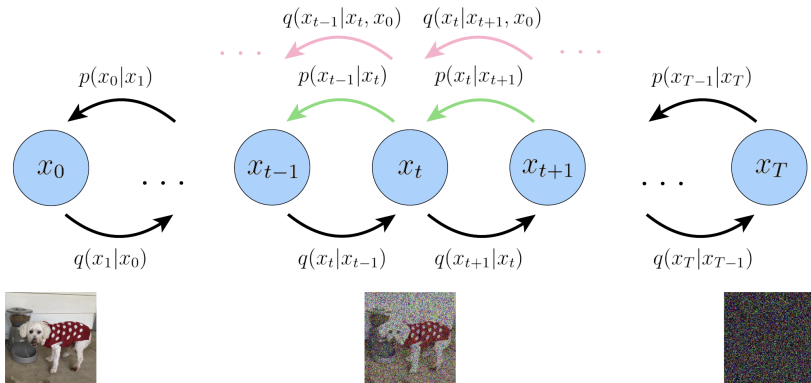
Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
 - Classifier Guidance
 - Classifier-Free Guidance

Outline

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ELBO for Gaussian Diffusion Model



$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

ELBO for Gaussian Diffusion Model

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Let's assume that

$$\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I} \quad \Rightarrow \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}).$$

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

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Theoretically, the optimal $\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t)$ lies in $[\tilde{\beta}_t, \beta_t]$:

- ▶ β_t is optimal for $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$;
- ▶ $\tilde{\beta}_t$ is optimal for $\mathbf{x}_0 \sim \delta(\mathbf{x}_0 - \mathbf{x}^*)$.

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

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ELBO for Gaussian Diffusion Model

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$$\begin{aligned} \mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(\mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \| \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I})) \\ &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t)\|^2 \right] \end{aligned}$$

ELBO for Gaussian Diffusion Model

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
3. Compute the ELBO:

$$\begin{aligned} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta, t}(\mathbf{x}_t)\|^2 \right]}_{\mathcal{L}_t}. \end{aligned}$$

ELBO for Gaussian Diffusion Model

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Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise $\mathbf{x}_{t-1} = \mu_{\theta, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

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Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

Reparametrization of DDPM

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$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

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Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

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- ▶ There is a linear relationship between $\boldsymbol{\epsilon}$, $\mathbf{x}_t$, and $\mathbf{x}_0$.
- ▶ Let's try to rewrite this mean using only $\mathbf{x}_t$ and $\boldsymbol{\epsilon}$.

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \boldsymbol{\epsilon}) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}} \right)$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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- ▶ Let's try to rewrite this mean using only $\mathbf{x}_t$ and $\boldsymbol{\epsilon}$.

$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \boldsymbol{\epsilon}) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}} \right) \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \cdot \boldsymbol{\epsilon} \end{aligned}$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}$$

Reparametrization of DDPM

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Reparametrization

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t)\end{aligned}$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon$$

$$\mu_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t)$$

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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At every step of the reverse process, we attempt to predict the noise ϵ that was used in the forward diffusion process!

Reparametrization of DDPM

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

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Let's drop the scaling coefficient.

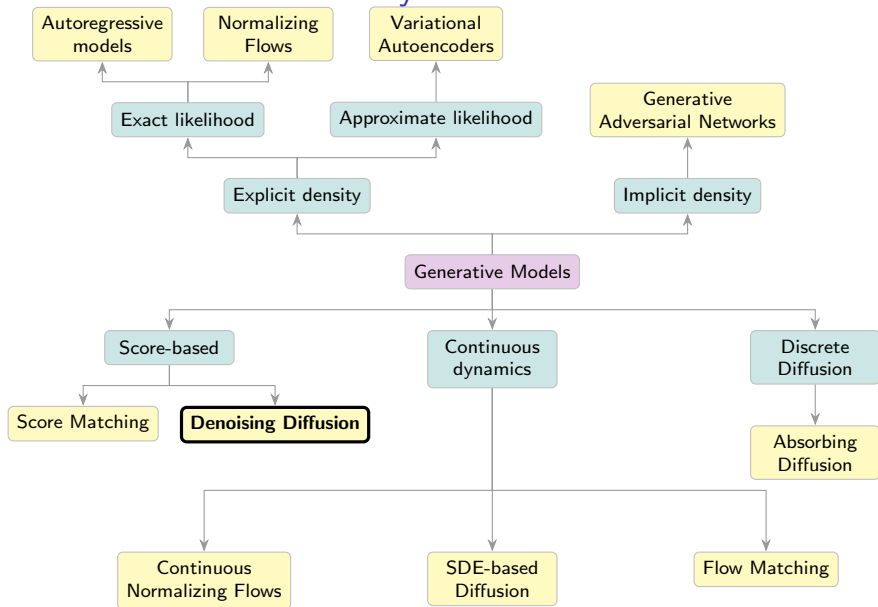
Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{2, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon) \right\|^2$$

Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
 - Classifier Guidance
 - Classifier-Free Guidance

Generative Models Taxonomy



Denoising Diffusion Probabilistic Model (DDPM)

DDPM is a VAE Model

- ▶ The encoder is a fixed Gaussian Markov chain $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$.
- ▶ The latent variable is hierarchical (at each step, its dimension equals the input's).
- ▶ The decoder is a simple Gaussian model $p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)$.
- ▶ The prior distribution is given by a parametric Gaussian Markov chain $p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

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Forward Process

1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$;
2. $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$;
3. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$.

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Reverse Process

1. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$;
2. $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$;
3. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.

Denoising Diffusion Probabilistic Model (DDPM)

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample $t \sim U\{1, T\}$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
3. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
4. Compute loss $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$.

Denoising Diffusion Probabilistic Model (DDPM)

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4. Compute loss $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\mu_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \mu_{\theta,t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[C_{1,t} \cdot \|\epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon\|_2^2 \right]$$

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We can reparameterize the model as:

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t) = -\frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t).$$

Denoising Diffusion as a Score-Based Generative Model

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We can reparameterize the model as:

$$\begin{aligned}\mathbf{s}_{\theta,t}(\mathbf{x}_t) &= -\frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t). \\ \mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[C_{2,t} \cdot \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]\end{aligned}$$

DDPM vs NCSN: Objectives

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

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$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

DDPM vs NCSN: Objectives

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$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[\mathbf{C}_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta,\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

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$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

Maximizing the ELBO leads to the same objective as denoising score matching!

DDPM vs NCSN: Sampling

DDPM Sampling (Ancestral Sampling)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mu_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

DDPM vs NCSN: Sampling

DDPM Sampling (Ancestral Sampling)

$$\begin{aligned}\mathbf{x}_T &\sim \mathcal{N}(0, \mathbf{I}) \\ \mathbf{x}_{t-1} &= \mu_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon\end{aligned}$$

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NCSN Sampling (Annealed Langevin Dynamics)

- ▶ Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
- ▶ Perform L steps of Langevin dynamics:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \cdot \mathbf{s}_{\theta,\sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \cdot \boldsymbol{\epsilon}_t^l.$$

- ▶ Set $\mathbf{x}_{t-1}^0 = \mathbf{x}_t^L$ and move to the next σ_t .

DDPM vs NCSN: Summary

Summary

- ▶ Different Markov chains:
 - ▶ DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
 - ▶ NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
 - ▶ One can generalize to $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$

Kingma D. et al. Variational Diffusion Models, 2021

Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

DDPM vs NCSN: Summary

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- ▶ The objectives coincide: ELBO $\equiv$ score-matching.

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 - ▶ One can generalize to $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- ▶ The objectives coincide: ELBO $\equiv$ score-matching.
- ▶ The sampling procedures differ:
 - ▶ Ancestral sampling in DDPM;
 - ▶ Annealed Langevin dynamics for NCSN;
 - ▶ Hybrid approaches that combine both updates are possible.

Kingma D. et al. Variational Diffusion Models, 2021

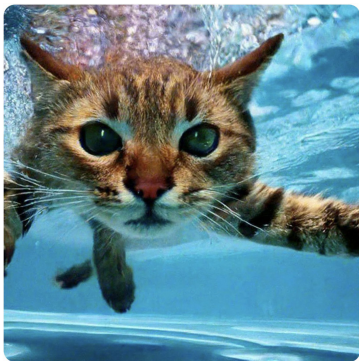
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Guidance

- ▶ Up to now, we have focused on **unconditional** generative models $p_{\theta}(\mathbf{x})$.
- ▶ In practice, most generative models are **conditional** (in the diffusion era it is called guided): $p_{\theta}(\mathbf{x}|\mathbf{y})$.
- ▶ Here, $\mathbf{y}$ might denote a class label or **text** (as in text-to-image tasks).



Кот ныряет в бассейн, как ребенок на обложке альбома Nevermind, реалистично



рука человека с пятью пальцами, ни четырьмя, ни шестью, а с 5 (пять) пальцами

Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution $p_{\text{data}}(\mathbf{x}|\mathbf{y})$).

- ▶ $\mathbf{y} = \emptyset$, $\mathbf{x} = \text{image}$ $\Rightarrow$ unconditional image model
- ▶ $\mathbf{y} = \text{class label}$, $\mathbf{x} = \text{image}$ $\Rightarrow$ class-conditional image model
- ▶ $\mathbf{y} = \text{text prompt}$, $\mathbf{x} = \text{image}$ $\Rightarrow$ text-to-image model
- ▶ $\mathbf{y} = \text{image}$, $\mathbf{x} = \text{image}$ $\Rightarrow$ image-to-image model
- ▶ $\mathbf{y} = \text{image}$, $\mathbf{x} = \text{text}$ $\Rightarrow$ image-to-text (image captioning) model
- ▶ $\mathbf{y} = \text{English text}$, $\mathbf{x} = \text{Russian text}$ $\Rightarrow$ sequence-to-sequence model (machine translation)
- ▶ $\mathbf{y} = \text{sound}$, $\mathbf{x} = \text{text}$ $\Rightarrow$ speech-to-text (automatic speech recognition) model
- ▶ $\mathbf{y} = \text{text}$, $\mathbf{x} = \text{sound}$ $\Rightarrow$ text-to-speech model

Label Guidance

Label: Ostrich (10th ImageNet class)



VQ-VAE (Proposed)

BigGAN deep

Text Guidance

Prompt: a stained glass window of a panda eating bamboo

Left: $\gamma = 1$, Right: $\gamma = 3$.



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- ▶ $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ for DDPMs.

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- ▶ $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ for DDPMs.

Challenge

- ▶ Empirically, images sampled with this procedure do not fit well enough to the desired label $\mathbf{y}$.
- ▶ Being able to control the strength of guidance is especially valuable.

Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
 - Classifier Guidance
 - Classifier-Free Guidance

Classifier Guidance

DDPM Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Generate the denoised image (unconditional generation):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

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Guided Generation

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What is the link between $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)$ and $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y})$?

Classifier Guidance: Guided Score Function

Guided Generation

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Classifier Guidance: Guidance Scale

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- ▶ Let us assume $\mathbf{y}$ is a class label.
- ▶ $p(\mathbf{y}|\mathbf{x}_t)$ is a classifier for noisy inputs.
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It is a natural idea to scale up the contribution of the guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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- ▶ The **guidance scale** γ adjusts the strength of classifier guidance.
- ▶ $\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y})$ is not the true guided score function $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Classifier Guidance: Distribution Sharpening

Scaled Guided Score Function

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Note: Increasing γ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Classifier Guidance: Overview

Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

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Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

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Classifier-Free Guidance

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Bayes theorem

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Scaled Guided Score Function

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$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Bayes theorem

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\theta}(\mathbf{x}_t|\mathbf{y})p(\mathbf{y})}{p_{\theta}(\mathbf{x}_t)} \right) \\ &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) \end{aligned}$$

Scaled Guided Score Function

$$\begin{aligned} \nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \\ &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot (\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)) = \\ &= (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) \end{aligned}$$

Classifier-Free Guidance: Formulation

Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$$

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$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

Classifier-Free Guidance: Formulation

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$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$$

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Naive training approach

- ▶ Train an unguided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$.
- ▶ Train a guided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Classifier-Free Guidance: Formulation

Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$$

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Naive training approach

- ▶ Train an unguided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$.
- ▶ Train a guided score function model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.
- ▶ Use their convex combination at inference.

Guided Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

Classifier-Free Guidance: Formulation

Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^\gamma \log p_\theta(\mathbf{x}_t|\mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y})$$

$$\mathbf{s}_{\theta,t}^\gamma(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

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How to avoid training two separate score function models?

Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

CFG

1. Introduce the “absence of conditioning” label $\mathbf{y} = \emptyset$.
2. Identify the unguided score function $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$.
3. Train a single model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \emptyset$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ and $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

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4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ and $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

Summary

- ▶ At each step, DDPM predicts the noise that was injected in the forward process.
- ▶ DDPMs are quite slow, since the model must be applied T times for sampling.
- ▶ DDPM and NCSN are intimately connected at the objective level.
- ▶ Guidance technique allows to make controllable generation via conditioning on labels or text prompts.
- ▶ Classifier guidance turns an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- ▶ Classifier-free guidance removes the need for such a classifier, yielding a practical recipe now widely used.